

Asymptotic Transseries of the Butcher–Pólya Rooted-Tree Numbers

Bell-Polynomial Coefficients, Lambert– W Reversion, and Exponentially Small Singularity Sectors

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Abstract

Let a_n denote OEIS A000081, the number of unlabelled rooted non-plane trees with n vertices, also called Pólya trees and, in the numerical- analysis literature, the unlabelled rooted trees underlying Butcher series. Its ordinary generating function $T(z) = \sum_{n \geq 1} a_n z^n$ satisfies

$$T(z) = z \exp \left(\sum_{k \geq 1} \frac{T(z^k)}{k} \right).$$

This article develops an all-orders asymptotic and transseries calculus for a_n and for its large-value functional inverse. The dominant singularity ρ is put into the Cayley normal form $T = \mathcal{C} \circ \zeta$, and every Puiseux coefficient at $z = \rho$ is expressed by a finite ordinary-Bell- polynomial formula in the derivatives of ζ . Exact coefficient transfer is then expanded by Bernoulli polynomials and complete exponential Bell polynomials. The resulting compact master formula is

$$a_n \sim \frac{\rho^{-n}}{\sqrt{\pi} n^{3/2}} \sum_{m \geq 0} \frac{\tau_m}{n^m}, \quad \tau_m = \sum_{j=0}^m \frac{(-1)^{j+1} (2j+1)!!}{2^{j+1}} t_{2j+1} g_{m-j} \left(j + \frac{1}{2} \right),$$

where the t_r are the local Puiseux coefficients and the $g_r(\beta)$ are given explicitly by complete Bell polynomials in Bernoulli-polynomial increments. Thus all asymptotic coefficients are obtained by a finite, triangular, auditable calculus.

Writing the dominant model as $C e^{\lambda x} x^{-3/2} S(x^{-1})$, its exact carrier inverse is a real W_{-1} expression. If $H = \log S$, the complete algebraic correction is $\nu(y) = x_0(y) + \sum_{m \geq 1} d_m x_0(y)^{-m}$, where the d_m obey an explicit triangular Bell-polynomial recurrence. An equivalent polynomial-logarithmic form $\nu(y) = X + q \log X + \sum_{m \geq 1} P_m(\log X) X^{-m}$ is derived, again with a general coefficient formula. Finally, a singular-germ recursion produces the beyond-all-orders sectors associated with inherited root singularities and any additional critical preimages; a corresponding carrier-first implicit recursion gives the inverse transseries. We distinguish rigorously between the smooth functional inverse, the stepwise inverse of the integer sequence, and the multiplicative reciprocal sequence.

Main result

With

$$\rho = 0.33832185689920769519611262571701705318377460753296779557 \dots,$$

$$\alpha = \rho^{-1} = 2.95576528565199497471481752412319458837549230466359659535 \dots,$$

and

$$C = 0.43992401257102530404090339143454476479808540794011985765 \dots,$$

the first normalized terms are

$$a_n \sim C\alpha^n n^{-3/2} \left(1 + \frac{0.1004034800964014348}{n} + \frac{0.5039343151023035331}{n^2} + \frac{1.972284908382278469}{n^3} + \frac{10.51734090413157399}{n^4} + \dots \right).$$

All displayed decimals are consequences of the exact coefficient formulae, not definitions of the coefficients.

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1 Introduction and statement of scope

1.1 The sequence and its three meanings of “inverse”

The sequence considered here is

$$0, 1, 1, 2, 4, 9, 20, 48, 115, 286, 719, 1842, 4766, \dots,$$

with $a_0 = 0$ and a_n equal to the number of isomorphism classes of rooted, unlabelled, non-plane trees on n vertices. Pólya’s multiset construction makes these objects especially natural in enumerative combinatorics. The same trees index elementary differentials and coefficients in Butcher B -series, which explains the name *Butcher tree sequence* used in numerical analysis [1, 7, 8].

The word *inverse* can refer to three different constructions, and a correct asymptotic treatment must separate them.

1. The *smooth functional inverse* $x = \nu(y)$ solves an asymptotic interpolation $\mathfrak{a}(x) = y$ for $x \rightarrow +\infty$. This is the main inverse developed in this article.
2. The *discrete generalized inverse*

$$N(y) := \min\{n \geq 1 : a_n \geq y\}$$

is a staircase. Its bounded lattice oscillation cannot be represented by an ordinary descending power series; nevertheless its complete unbounded and vanishing asymptotic content is inherited from $\nu(y)$.

3. The *multiplicative reciprocal* $1/a_n$ is not a functional inverse. Its all-orders expansion is included in [Section 8](#) because it follows at essentially no extra cost from the same coefficient calculus.

1.2 What “full transseries” means here

There are two distinct levels of completeness.

First, the dominant singularity at $z = \rho$ gives a rigorous Poincaré expansion to arbitrary algebraic order,

$$a_n = \frac{\rho^{-n}}{\sqrt{\pi} n^{3/2}} \left(\sum_{m=0}^{M-1} \frac{\tau_m}{n^m} + \mathcal{O}(n^{-M}) \right) \quad (M \geq 1 \text{ fixed}).$$

The main algebraic contribution of this article is a simpler coefficient formula for τ_m than a direct repeated singularity-analysis calculation: it is a finite convolution of the odd Puiseux coefficients with universal gamma-ratio coefficients.

Second, analytic continuation of the recursive generating equation produces further singular germs. Each germ gives a different exponential weight σ^{-n} , so the coefficient sequence has a multi-sector expansion ordered by $|\sigma|$. The inherited family $\sigma^m = \rho$ produces the characteristic scales $\rho^{-n/m}$ with root-of-unity phases. Additional solutions of the critical equation $\zeta(z) = 1/e$ may also occur on continued sheets. We give a complete local coefficient recursion and a finite-radius transfer theorem under explicit continuation and separation hypotheses. This is the appropriate rigorous meaning of a beyond-all-orders transseries: every finite collection of accessible singularities can be expanded and transferred, while a global assertion that the recursively generated list is exhaustive would require a separate theorem on the full Riemann surface.

A point requiring care

The dominant all-orders expansion and its algebraic functional inverse are unconditional within standard singularity-analysis hypotheses known for Pólya trees. The global exponential completion is stated as a finite-level theorem under explicit analytic-continuation hypotheses. This distinction prevents a formal singularity grammar from being mistaken for an unproved global classification of every sheet and every critical coincidence.

1.3 Relation to the supplied coefficient and transseries calculi

The derivations are organized in the style of the supplied ProveIt drafts: ordinary Bell polynomials encode powers of a series, complete exponential Bell polynomials encode exponentials and Faà di Bruno composition, Bernoulli polynomials encode gamma-ratio transfer, and inversion is performed only after the dominant carrier has been normalized [2, 12, 13]. Formal identities are kept separate from analytic remainder statements. This makes the output suitable both for symbolic implementation and for later formalization.

1.4 Road map

Section 2 fixes the coefficient conventions. Section 3 derives the exact generating equations and recurrence. The Cayley normal form and the universal square-root coefficients are developed in Section 4. Pólya-specific Puiseux coefficients are obtained in Section 5. Section 6 proves the master formula for τ_m and records numerical coefficients. The logarithmic normal form, reciprocal sequence, and ratio consequences are in Sections 7 and 8. The Lambert carrier and the two complete forms of the functional inverse occupy Sections 9 and 10. The discrete inverse is treated in Section 11. The singularity transseries and its inverse are developed in Sections 12 and 13. Algorithms, validation, and reproducibility checks conclude the article.

2 Coefficient calculi and notation

2.1 Ordinary partial Bell polynomials

Definition 2.1 (Ordinary partial Bell polynomials). For indeterminates $x_1, x_2, \dots$, define

$$\left(\sum_{j \geq 1} x_j u^j \right)^k = \sum_{n \geq k} \hat{B}_{n,k}(x_1, x_2, \dots) u^n,$$

with $\hat{B}_{0,0} = 1$ and $\hat{B}_{n,0} = 0$ for $n > 0$. Equivalently,

$$\hat{B}_{n,k}(x_1, x_2, \dots) = \sum_{\substack{j_1 + j_2 + \dots = k \\ j_1 + 2j_2 + 3j_3 + \dots = n}} \frac{k!}{j_1! j_2! \dots} \prod_{m \geq 1} x_m^{j_m}. \quad (2.1)$$

Thus, for arbitrary γ and $U(u) = \sum_{j \geq 1} x_j u^j$,

$$[u^n](1 + U(u))^\gamma = \sum_{k=0}^n \binom{\gamma}{k} \hat{B}_{n,k}(x_1, x_2, \dots). \quad (2.2)$$

This single identity is the workhorse for fractional powers in the Puiseux calculus and for negative powers in inverse-series calculations.

2.2 Exponential Bell polynomials

Definition 2.2 (Partial and complete exponential Bell polynomials). The partial exponential Bell polynomial $B_{n,k}$ is defined by

$$\frac{1}{k!} \left(\sum_{j \geq 1} x_j \frac{u^j}{j!} \right)^k = \sum_{n \geq k} B_{n,k}(x_1, x_2, \dots) \frac{u^n}{n!}.$$

The complete exponential Bell polynomial is

$$Y_n(x_1, \dots, x_n) = \sum_{k=0}^n B_{n,k}(x_1, \dots, x_{n-k+1}).$$

It is characterized by

$$\exp \left(\sum_{j \geq 1} x_j \frac{u^j}{j!} \right) = \sum_{n \geq 0} Y_n(x_1, \dots, x_n) \frac{u^n}{n!}. \quad (2.3)$$

The Faà di Bruno formula is then

$$\frac{d^n}{dz^n} f(g(z)) = \sum_{k=1}^n f^{(k)}(g(z)) B_{n,k}(g'(z), g''(z), \dots, g^{(n-k+1)}(z)). \quad (2.4)$$

2.3 Bernoulli polynomials and gamma-ratio coefficients

Let the Bernoulli polynomials be defined by

$$\frac{te^{xt}}{e^t - 1} = \sum_{n \geq 0} B_n(x) \frac{t^n}{n!}.$$

For $\beta \in \mathbb{C}$ define

$$\kappa_m(\beta) := \frac{(-1)^{m+1}}{m(m+1)} (B_{m+1}(-\beta) - B_{m+1}(1)), \quad m \geq 1, \quad (2.5)$$

and define $g_r(\beta)$ by

$$\exp \left(\sum_{m \geq 1} \kappa_m(\beta) u^m \right) = \sum_{r \geq 0} g_r(\beta) u^r. \quad (2.6)$$

Thus

$$g_r(\beta) = \frac{1}{r!} Y_r(1! \kappa_1(\beta), 2! \kappa_2(\beta), \dots, r! \kappa_r(\beta)), \quad (2.7)$$

or, computationally,

$$g_0(\beta) = 1, \quad g_r(\beta) = \frac{1}{r} \sum_{m=1}^r m \kappa_m(\beta) g_{r-m}(\beta). \quad (2.8)$$

The first values are

$$\begin{aligned} g_0(\beta) &= 1, \\ g_1(\beta) &= \frac{\beta(\beta+1)}{2}, \\ g_2(\beta) &= \frac{\beta(\beta+1)(\beta+2)(3\beta+1)}{24}, \\ g_3(\beta) &= \frac{\beta^2(\beta+1)^2(\beta+2)(\beta+3)}{48}, \\ g_4(\beta) &= \frac{\beta(\beta+1)(\beta+2)(\beta+3)(\beta+4)(15\beta^3+30\beta^2+5\beta-2)}{5760}. \end{aligned}$$

Lemma 2.3 (Gamma-ratio expansion). *For fixed β and $n \rightarrow \infty$ in a sector avoiding the negative real axis,*

$$\frac{\Gamma(n - \beta)}{\Gamma(n + 1)} \sim n^{-\beta-1} \sum_{r \geq 0} \frac{g_r(\beta)}{n^r}. \quad (2.9)$$

Proof. The standard Bernoulli-polynomial expansion of the logarithm of a gamma ratio gives

$$\log \frac{\Gamma(n - \beta)}{\Gamma(n + 1)} \sim -(\beta + 1) \log n + \sum_{m \geq 1} \frac{\kappa_m(\beta)}{n^m}.$$

Exponentiating and applying (2.3) gives (2.9), with the coefficient descriptions (2.7) and (2.8). $\square$

Section summary

There are only three coefficient mechanisms in the dominant calculation: $\widehat{B}_{n,k}$ for powers of ordinary series, $B_{n,k}$ and Y_n for composition and exponentiation, and $g_r(\beta)$ for exact singularity transfer. Every later formula reduces to these finite polynomial objects.

3 Exact combinatorial and generating-function equations

3.1 Pólya's multiset equation

A rooted non-plane tree consists of a root vertex carrying an unordered multiset of rooted non-plane trees. The unlabelled multiset construction and Pólya cycle-index substitution therefore give [11, 5, 8]

$$T(z) = z \exp \left(\sum_{k \geq 1} \frac{T(z^k)}{k} \right), \quad T(z) = \sum_{n \geq 1} a_n z^n. \quad (3.1)$$

Separating the $k = 1$ term yields

$$T(z) = \zeta(z) e^{T(z)}, \quad \zeta(z) := z \exp R(z), \quad R(z) := \sum_{k \geq 2} \frac{T(z^k)}{k}. \quad (3.2)$$

The Euler-product form follows by expanding the logarithm:

$$T(z) = \frac{z}{\prod_{m \geq 1} (1 - z^m)^{a_m}}. \quad (3.3)$$

3.2 The exact quadratic-time recurrence

Define the divisor transform

$$b_k := \sum_{d|k} d a_d. \quad (3.4)$$

Logarithmic differentiation of (3.3) gives

$$\frac{zT'(z)}{T(z)} = 1 + \sum_{k \geq 1} b_k z^k.$$

Coefficient extraction proves the following recurrence.

Proposition 3.1 (Exact recurrence). *With $a_0 = 0$, $a_1 = 1$, one has, for $n \geq 2$,*

$$a_n = \frac{1}{n-1} \sum_{k=1}^{n-1} \left(\sum_{d|k} d a_d \right) a_{n-k} = \frac{1}{n-1} \sum_{k=1}^{n-1} b_k a_{n-k}. \quad (3.5)$$

In particular, the first N values require $\mathcal{O}(N^2)$ arithmetic operations when the divisor transform is updated by a sieve.

Proof. Multiplying the logarithmic-derivative identity by $T(z)$ gives

$$zT'(z) = T(z) + T(z) \sum_{k \geq 1} b_k z^k.$$

The coefficient of z^n is $na_n = a_n + \sum_{k=1}^{n-1} b_k a_{n-k}$, which is (3.5). For the complexity statement, after computing a_d add da_d to $b_d, b_{2d}, b_{3d}, \dots, b_{\lfloor N/d \rfloor d}$. The sieve costs $\mathcal{O}(N \log N)$ additions, and the convolution in (3.5) costs $\mathcal{O}(N^2)$ arithmetic operations. $\square$

3.3 Critical characterization of the dominant singularity

Let ρ be the radius of convergence of T . Since the coefficients are nonnegative, Pringsheim's theorem puts a singularity at the positive point $z = \rho$. For $|z| < \sqrt{\rho}$, every z^k with $k \geq 2$ lies in the disc of convergence of T , so R and ζ are analytic there. The implicit equation $F(z, w) = w - \zeta(z)e^w = 0$ becomes singular when

$$F = 0, \quad F_w = 1 - \zeta(z)e^w = 0.$$

Consequently the positive critical point satisfies

$$T(\rho) = 1, \quad \zeta(\rho) = e^{-1}, \quad (3.6)$$

or equivalently

$$\log \rho + 1 + \sum_{k \geq 2} \frac{T(\rho^k)}{k} = 0. \quad (3.7)$$

The left side is strictly increasing on the relevant positive interval, so this equation characterizes ρ uniquely [10, 5, 6].

Proposition 3.2 (Numerical constants). *Solving (3.7) with the exact recurrence (3.5) gives*

$$\rho = 0.33832185689920769519611262571701705318377460753296779557 \dots, \quad (3.8)$$

$$\alpha := \rho^{-1} = 2.95576528565199497471481752412319458837549230466359659535 \dots, \quad (3.9)$$

$$\lambda := \log \alpha = 1.08375759724462466048271198089617427904966212420267223491 \dots \quad (3.10)$$

Numerical certification strategy. For any trial r near ρ , every argument r^k , $k \geq 2$, lies far inside the disc $|z| < \rho$. Hence $T(r^k) = \sum_{n \geq 1} a_n r^{kn}$ converges geometrically, and the omitted tail can be bounded by a geometric majorant after finitely many exact coefficients have been generated. Interval evaluation of (3.7) and its positive derivative then encloses the unique root. The displayed digits agree with the established Otter growth constant α [9]. $\square$

4 Cayley normal form and the universal square-root germ

4.1 The Cayley tree function

Let $\mathcal{C}(w)$ be the solution analytic at the origin of

$$\mathcal{C}(w) = we^{\mathcal{C}(w)}, \quad \mathcal{C}(0) = 0. \quad (4.1)$$

Equivalently,

$$\mathcal{C}(w) = -W_0(-w). \quad (4.2)$$

The first critical point is $(w, \mathcal{C}) = (e^{-1}, 1)$. By (3.2), the Pólya-tree generating function is exactly

$$T(z) = \mathcal{C}(\zeta(z)) \quad (4.3)$$

throughout the common domain of the branches selected at the origin.

To obtain the complete local germ of $\mathcal{C}$, put

$$w = \frac{1-u}{e}, \quad \mathcal{C}(w) = 1-v. \quad (4.4)$$

Substitution in (4.1) gives

$$u = 1 - (1-v)e^v = \frac{v^2}{2}\Phi(v), \quad (4.5)$$

where

$$\Phi(v) := \frac{2(1 - (1-v)e^v)}{v^2} = \sum_{m \geq 0} \frac{2(m+1)}{(m+2)!} v^m = 1 + \frac{2}{3}v + \frac{1}{4}v^2 + \dots \quad (4.6)$$

Theorem 4.1 (Universal Cayley Puiseux coefficients). *On the branch reached from $0 < w < e^{-1}$,*

$$\mathcal{C}\left(\frac{1-u}{e}\right) = 1 + \sum_{r \geq 1} c_r u^{r/2}, \quad (4.7)$$

where, for every $r \geq 1$,

$$c_r = -\frac{2^{r/2}}{r} [v^{r-1}] \Phi(v)^{-r/2}. \quad (4.8)$$

Equivalently, writing $\phi_m = 2(m+1)/(m+2)!$ for $m \geq 1$,

$$c_r = -\frac{2^{r/2}}{r} \sum_{k=0}^{r-1} \binom{-r/2}{k} \widehat{B}_{r-1,k}(\phi_1, \phi_2, \dots). \quad (4.9)$$

Proof. Set $s = u^{1/2}$ with the positive square root for $u > 0$. Equation (4.5) is equivalent to

$$v = s \Psi(v), \quad \Psi(v) = \sqrt{2} \Phi(v)^{-1/2},$$

with $\Psi(0) = \sqrt{2} \neq 0$. The Lagrange–Bürmann formula gives

$$[s^r]v(s) = \frac{1}{r} [v^{r-1}] \Psi(v)^r = \frac{2^{r/2}}{r} [v^{r-1}] \Phi(v)^{-r/2}.$$

Since $\mathcal{C} = 1 - v$, the coefficient in (4.7) is the negative of this number, proving (4.8). Formula (4.9) is (2.2) applied to $\Phi(v) = 1 + \sum_{m \geq 1} \phi_m v^m$. $\square$

The first coefficients are

$$\begin{aligned} c_1 &= -\sqrt{2}, & c_2 &= \frac{2}{3}, & c_3 &= -\frac{11\sqrt{2}}{36}, & c_4 &= \frac{43}{135}, \\ c_5 &= -\frac{769\sqrt{2}}{4320}, & c_6 &= \frac{1768}{8505}, & c_7 &= -\frac{680863\sqrt{2}}{5443200}, & c_8 &= \frac{3926}{25515}. \end{aligned} \quad (4.10)$$

These constants are universal: the particular tree family enters only through the analytic change of variable $u = 1 - e\zeta(z)$.

4.2 Why the square root is unavoidable

The critical equations for $F(w, y) = y - we^y$ are $F(e^{-1}, 1) = F_y(e^{-1}, 1) = 0$, whereas $F_{yy}(e^{-1}, 1) = -1 \neq 0$ and $F_w(e^{-1}, 1) = -e \neq 0$. The Weierstrass preparation theorem therefore gives a quadratic local branching. The Lagrange formula above does more than establish the exponent: it gives every coefficient in the branch germ by one finite coefficient extraction.

Structural checkpoint

All family dependence is now isolated in $q(s) := 1 - e\zeta(\rho(1 - s))$. The map $q \mapsto T$ is the universal composition $1 + \sum_{r \geq 1} c_r q^{r/2}$. This carrier-first normalization is the key simplification behind the later all-orders coefficient formula.

5 All Puiseux coefficients at the Otter singularity

5.1 The analytic perturbation series

Put

$$s := 1 - \frac{z}{\rho} \quad (5.1)$$

and define the analytic series

$$q(s) := 1 - e\zeta(\rho(1 - s)) = \sum_{m \geq 1} q_m s^m. \quad (5.2)$$

The constant term vanishes by (3.6). Taylor's formula gives

$$q_m = \frac{(-1)^{m+1} e \rho^m}{m!} \zeta^{(m)}(\rho), \quad m \geq 1. \quad (5.3)$$

In particular $q_1 = e\rho\zeta'(\rho) > 0$.

Factor

$$q(s) = q_1 s (1 + U(s)), \quad U(s) = \sum_{j \geq 1} u_j s^j, \quad u_j := \frac{q_{j+1}}{q_1}. \quad (5.4)$$

The desired local expansion is

$$T(z) = 1 + \sum_{n \geq 1} t_n \left(1 - \frac{z}{\rho}\right)^{n/2} = 1 + \sum_{n \geq 1} t_n s^{n/2}. \quad (5.5)$$

Theorem 5.1 (Finite Bell formula for every t_n). *Let c_r be given by (4.8). Then*

$$t_n = \sum_{\substack{1 \leq r \leq n \\ r \equiv n \pmod{2}}} c_r q_1^{r/2} [s^{(n-r)/2}] (1 + U(s))^{r/2}. \quad (5.6)$$

Equivalently,

$$t_n = \sum_{\substack{1 \leq r \leq n \\ r \equiv n \pmod{2}}} c_r q_1^{r/2} \sum_{k=0}^{(n-r)/2} \binom{r/2}{k} \widehat{B}_{(n-r)/2,k}(u_1, u_2, \dots). \quad (5.7)$$

Thus t_n is a finite Laurent-polynomial expression in $q_1^{1/2}$, polynomial in $q_2, \dots, q_{(n+1)/2}$, with coefficients generated by the universal rational-radical constants $c_1, \dots, c_n$.

Proof. By (4.3), (4.7), and (5.2),

$$T(z) = 1 + \sum_{r \geq 1} c_r q(s)^{r/2} = 1 + \sum_{r \geq 1} c_r q_1^{r/2} s^{r/2} (1 + U(s))^{r/2}.$$

A contribution to $s^{n/2}$ is possible exactly when $n - r$ is a nonnegative even integer. Extracting the remaining integral power of s gives (5.6); inserting (2.2) gives (5.7). $\square$

The first three nontrivial consequences are

$$t_1 = -\sqrt{2q_1}, \quad (5.8)$$

$$t_2 = \frac{2}{3}q_1, \quad (5.9)$$

$$t_3 = -\frac{11\sqrt{2}}{36}q_1^{3/2} - \frac{\sqrt{2}}{2}q_2q_1^{-1/2}. \quad (5.10)$$

Formula (5.7) is preferable to a rapidly growing list of hand-expanded expressions because it preserves the triangular dependence and is directly implementable.

5.2 Derivatives of ζ from subcritical data

The remaining task is to compute the derivatives in (5.3). Let

$$L(z) := \log \zeta(z) = \log z + R(z), \quad R(z) = \sum_{k \geq 2} \frac{T(z^k)}{k}. \quad (5.11)$$

Set

$$\ell_m := L^{(m)}(\rho) = \frac{(-1)^{m-1}(m-1)!}{\rho^m} + R^{(m)}(\rho). \quad (5.12)$$

Then the complete Bell polynomial gives

$$\zeta^{(m)}(\rho) = e^{-1} Y_m(\ell_1, \ell_2, \dots, \ell_m). \quad (5.13)$$

For $g_k(z) = z^k$, Faà di Bruno yields

$$R^{(m)}(\rho) = \sum_{k \geq 2} \frac{1}{k} \sum_{r=1}^m T^{(r)}(\rho^k) B_{m,r}((k)_1 \rho^{k-1}, (k)_2 \rho^{k-2}, \dots, (k)_{m-r+1} \rho^{k-m+r-1}). \quad (5.14)$$

Here $(k)_j = k(k-1) \cdots (k-j+1)$, with the convention that it is zero for $j > k$. Every argument ρ^k , $k \geq 2$, is strictly subcritical, so

$$T^{(r)}(\rho^k) = \sum_{n \geq r} (n)_r a_n \rho^{k(n-r)} \quad (5.15)$$

converges geometrically. Equations (5.3), (5.13), and (5.14) therefore constitute a complete and numerically stable all-orders prescription for the input q_m .

Remark 5.2 (A direct coefficient alternative). For numerical work it is often faster to expand $R(\rho(1-s))$ directly. Since

$$T(\rho^k(1-s)^k) = \sum_{n \geq 1} a_n \rho^{kn} (1-s)^{kn},$$

one has

$$[s^m]R(\rho(1-s)) = (-1)^m \sum_{k \geq 2} \frac{1}{k} \sum_{n \geq 1} a_n \rho^{kn} \binom{kn}{m}. \quad (5.16)$$

The double sum converges geometrically. Exponentiating this ordinary power series by complete Bell polynomials recovers the same q_m without explicit derivatives.

5.3 Numerical Puiseux coefficients

The first coefficients obtained from the preceding formulas are shown in [Table 1](#); they agree with the independent table in [\[6\]](#). Even-indexed coefficients belong to the analytic part of the local germ; odd-indexed coefficients create the algebraic coefficient asymptotics.

Table 1: Puiseux coefficients in $T(z) = 1 + \sum_{n \geq 1} t_n (1 - z/\rho)^{n/2}$.

n	t_n	n	t_n
1	-1.5594900203746408855	10	0.2059848312779074187
2	0.8106697078826992814	11	-0.05272470849819056232
3	-0.2854870216128456053	12	0.01702656875495366945
4	0.1653723657120838935	13	-0.1523706243663253963
5	-0.3424599704021542010	14	0.1737028832998504629
6	0.3174072259465285636	15	-0.01447370373952704486
7	-0.1077788002916310092	16	-0.02189951761121556223
8	0.06138495705583510469	17	-0.1445471935709097055
9	-0.1952123835975564636	18	0.1760771088850177791

Section summary

The universal coefficients c_r are produced once by (4.8). The family-specific derivatives of ζ are produced by (5.13)–(5.14). Their composition is the finite Bell sum (5.7). No coefficient requires ad hoc reversion or numerical fitting.

6 Coefficient transfer and the all-orders dominant expansion

6.1 Exact transfer of one singular monomial

For $\beta \notin \mathbb{N}_0$, the binomial identity gives the exact coefficient formula

$$[z^n] \left(1 - \frac{z}{\rho}\right)^\beta = \rho^{-n} \frac{\Gamma(n - \beta)}{\Gamma(-\beta)\Gamma(n + 1)}. \quad (6.1)$$

Combining (6.1) with [Theorem 2.3](#) yields

$$[z^n] \left(1 - \frac{z}{\rho}\right)^\beta \sim \frac{\rho^{-n} n^{-\beta-1}}{\Gamma(-\beta)} \sum_{r \geq 0} \frac{g_r(\beta)}{n^r}. \quad (6.2)$$

For half-integers,

$$\frac{1}{\Gamma(-j - \frac{1}{2})} = \frac{(-1)^{j+1}(2j+1)!!}{2^{j+1}\sqrt{\pi}}, \quad j \geq 0. \quad (6.3)$$

The even terms $t_{2j}(1 - z/\rho)^j$ in the local expansion are analytic polynomials and do not contribute to the large- n coefficient tail. The odd terms contain all orders of the dominant algebraic asymptotics.

6.2 The master coefficient theorem

Theorem 6.1 (Full dominant expansion). *For each fixed integer $M \geq 1$,*

$$a_n = \frac{\rho^{-n}}{\sqrt{\pi} n^{3/2}} \left(\sum_{m=0}^{M-1} \frac{\tau_m}{n^m} + \mathcal{O}(n^{-M}) \right), \quad n \rightarrow \infty, \quad (6.4)$$

where

$$\tau_m = \sum_{j=0}^m D_j t_{2j+1} g_{m-j} \left(j + \frac{1}{2} \right), \quad D_j := \frac{(-1)^{j+1}(2j+1)!!}{2^{j+1}}. \quad (6.5)$$

Equivalently,

$$\tau_m = \sqrt{\pi} \sum_{j=0}^m \frac{t_{2j+1}}{\Gamma(-j - \frac{1}{2})} g_{m-j} \left(j + \frac{1}{2} \right). \quad (6.6)$$

Together with (4.9), (5.7), (5.3), (5.13), and (5.14), this is a finite general formula for every asymptotic coefficient τ_m .

Proof. The Pólya-tree generating function is analytic in a dented neighborhood of its unique dominant singularity ρ , and its local expansion is (5.5). Retain the odd terms $t_{2j+1}(1 - z/\rho)^{j+1/2}$ with $0 \leq j \leq M-1$ and absorb the remaining singular terms into an admissible remainder. Applying (6.2) term by term gives

$$[z^n] t_{2j+1} \left(1 - \frac{z}{\rho} \right)^{j+1/2} \sim t_{2j+1} \rho^{-n} n^{-j-3/2} \frac{1}{\Gamma(-j - \frac{1}{2})} \sum_{r \geq 0} \frac{g_r(j + \frac{1}{2})}{n^r}.$$

After factoring $\rho^{-n}/(\sqrt{\pi} n^{3/2})$, the coefficient of n^{-m} is obtained from pairs $j + r = m$. Inserting (6.3) gives (6.5). Standard transfer with a sufficiently long local expansion gives the stated remainder for every fixed M . $\square$

Corollary 6.2 (Otter's leading constant). *The leading coefficient is*

$$\tau_0 = -\frac{t_1}{2} = \sqrt{\frac{q_1}{2}}, \quad C := \frac{\tau_0}{\sqrt{\pi}} = \sqrt{\frac{e\rho\zeta'(\rho)}{2\pi}}. \quad (6.7)$$

Consequently

$$a_n \sim C \alpha^n n^{-3/2}, \quad (6.8)$$

where $\alpha = \rho^{-1}$ and

$$C = 0.43992401257102530404090339143454476479808540794011985765 \dots \quad (6.9)$$

Proof. Set $m = 0$ in (6.5), use $D_0 = -1/2$ and $g_0 = 1$, and then insert $t_1 = -\sqrt{2q_1}$ from (5.8) and $q_1 = e\rho\zeta'(\rho)$ from (5.3). $\square$

6.3 First coefficients in expanded form

The first five instances of (6.5) are

$$\tau_0 = -\frac{t_1}{2}, \quad (6.10)$$

$$\tau_1 = -\frac{3(t_1 - 4t_3)}{16}, \quad (6.11)$$

$$\tau_2 = -\frac{5(5t_1 - 72t_3 + 96t_5)}{256}, \quad (6.12)$$

$$\tau_3 = -\frac{105(t_1 - 44t_3 + 160t_5 - 128t_7)}{2048}, \quad (6.13)$$

$$\tau_4 = -\frac{21}{65536}(79t_1 - 10800t_3 + 81600t_5 - 161280t_7 + 92160t_9). \quad (6.14)$$

The compact convolution (6.5) scales much better than these fully expanded expressions.

6.4 Numerical coefficients

Table 2: Coefficients in $a_n \sim \rho^{-n}(\pi n^3)^{-1/2} \sum_{m \geq 0} \tau_m n^{-m}$.

m	τ_m	m	τ_m
0	0.7797450101873204428	10	$1.599693249293173313 \times 10^7$
1	0.07828911261061096206	11	$2.928225313353392231 \times 10^8$
2	0.3929402676631860185	12	$5.914523441293932519 \times 10^9$
3	1.537879315978838091	13	$1.305991927898971332 \times 10^{11}$
4	8.200844090435596159	14	$3.128498399789518761 \times 10^{12}$
5	57.29291473494343788	15	$8.078305401468884289 \times 10^{13}$
6	503.0445050262735818	16	$2.236301680891633732 \times 10^{15}$
7	5359.600933884326033	17	$6.605960869699189564 \times 10^{16}$
8	67342.06920114653048	18	$2.073828085209893749 \times 10^{18}$
9	975425.4970695924732		

It is often preferable to factor out the leading constant. Define

$$S(u) := 1 + \sum_{m \geq 1} s_m u^m, \quad s_m := \frac{\tau_m}{\tau_0}. \quad (6.15)$$

Then

$$\boxed{a_n \sim C \alpha^n n^{-3/2} S(n^{-1})}. \quad (6.16)$$

The first normalized coefficients are

$$\begin{aligned} s_1 &= 0.1004034800964014347637, \\ s_2 &= 0.5039343151023035331022, \\ s_3 &= 1.972284908382278469206, \\ s_4 &= 10.51734090413157399274, \\ s_5 &= 73.47647498402047106417, \\ s_6 &= 645.139755245661293 \dots, \\ s_7 &= 6873.530274463408 \dots, \\ s_8 &= 86364.21948371140 \dots \end{aligned}$$

6.5 Remainders, divergence, and optimal truncation

The statement “full expansion” means that every fixed truncation has the next Poincaré order; it does not assert convergence of $\sum \tau_m n^{-m}$. The rapid growth in Table 2 is consistent with a divergent, Gevrey-type expansion controlled by more distant singular structure. At a fixed moderately sized n , adding terms first improves the approximation and eventually worsens it. This is the usual least-term principle.

A useful diagnostic follows directly from the terms $|\tau_m|n^{-m}$. Truncate near the smallest available term, and estimate the algebraic truncation uncertainty by the first omitted term. A rigorous bound requires uniform control of the local remainder; the least-term rule is a numerical strategy rather than a substitute for that bound.

7 Logarithmic normal form and related sequence transforms

7.1 The logarithm of the asymptotic series

Define

$$H(u) := \log S(u) = \sum_{m \geq 1} h_m u^m. \quad (7.1)$$

By the ordinary Bell-polynomial convention,

$$h_m = \sum_{k=1}^m \frac{(-1)^{k-1}}{k} \widehat{B}_{m,k}(s_1, s_2, \dots). \quad (7.2)$$

Equivalently, differentiating $S = e^H$ gives the triangular recurrence

$$h_m = s_m - \frac{1}{m} \sum_{k=1}^{m-1} k h_k s_{m-k}. \quad (7.3)$$

Therefore

$$\log a_n \sim n\lambda - \frac{3}{2} \log n + \log C + \sum_{m \geq 1} \frac{h_m}{n^m}. \quad (7.4)$$

The first values are

$$\begin{aligned} h_1 &= 0.1004034800964014347637, \\ h_2 &= 0.4988938856945692935703, \\ h_3 &= 1.922025533841821821591, \\ h_4 &= 10.19739642337376887053, \\ h_5 &= 71.47146706207073604412, \\ h_6 &= 630.859937607337 \dots \end{aligned}$$

This logarithmic normal form is the natural input for functional inversion.

7.2 Ratio expansion and eventual monotonicity

Subtracting (7.4) at n and $n+1$ gives

$$\log \frac{a_{n+1}}{a_n} \sim \lambda - \frac{3}{2n} + \frac{\frac{3}{4} - h_1}{n^2} + \frac{-\frac{1}{2} + h_1 - 2h_2}{n^3} + \dots. \quad (7.5)$$

Exponentiation gives a complete expansion for a_{n+1}/a_n . In particular,

$$\frac{a_{n+1}}{a_n} \longrightarrow \alpha > 1, \quad (7.6)$$

so the sequence is strictly increasing for all sufficiently large n , and the generalized inverse in [Section 11](#) is well defined without any large- n ambiguity.

8 The multiplicative reciprocal sequence

Although the main inverse is functional, the reciprocal sequence has an immediate all-orders expansion. Define

$$R_*(u) := \frac{1}{S(u)} = \sum_{m \geq 0} r_m u^m, \quad r_0 = 1. \quad (8.1)$$

Then

$$\boxed{\frac{1}{a_n} \sim C^{-1} \rho^n n^{3/2} \sum_{m \geq 0} \frac{r_m}{n^m}.} \quad (8.2)$$

The coefficients have both a triangular and a Bell-polynomial form:

$$r_m = - \sum_{k=1}^m s_k r_{m-k}, \quad (8.3)$$

$$r_m = \sum_{k=1}^m (-1)^k \widehat{B}_{m,k}(s_1, s_2, \dots), \quad m \geq 1. \quad (8.4)$$

For example,

$$r_1 = -s_1, \quad r_2 = s_1^2 - s_2, \quad r_3 = -s_1^3 + 2s_1 s_2 - s_3.$$

This section also illustrates why it is essential not to call $1/a_n$ the functional inverse: its exponential scale is ρ^n , whereas the functional inverse grows only logarithmically in its argument.

9 Functional inversion: the exact Lambert carrier

9.1 The smooth asymptotic model

Set

$$p := \frac{3}{2} \quad (9.1)$$

and regard

$$\mathcal{A}(x) := C e^{\lambda x} x^{-p} S(x^{-1}) \quad (9.2)$$

as a formal asymptotic interpolation of a_n . For sufficiently large real x , every finite truncation is strictly increasing. We denote by $\nu(y)$ the large real solution of

$$\mathcal{A}(\nu(y)) = y, \quad (9.3)$$

interpreted asymptotically: substitution of an M -term approximation for S determines ν through the corresponding order.

Taking logarithms gives the canonical reversion problem

$$\log \frac{y}{C} = \lambda x - p \log x + H(x^{-1}). \quad (9.4)$$

The slowly varying factor H should not be expanded until the elementary exponential–power carrier has been inverted exactly.

9.2 Exact inversion of the exponential–power carrier

Let $x_0(y)$ solve

$$y = Ce^{\lambda x_0} x_0^{-p}. \quad (9.5)$$

Raising both sides to the power $-1/p$ gives

$$-\frac{\lambda}{p} \left(\frac{C}{y} \right)^{1/p} = -\frac{\lambda x_0}{p} e^{-\lambda x_0/p}.$$

Hence

$$x_0(y) = -\frac{p}{\lambda} W_{-1} \left(-\frac{\lambda}{p} \left(\frac{C}{y} \right)^{1/p} \right). \quad (9.6)$$

For the rooted-tree exponent $p = 3/2$,

$$x_0(y) = -\frac{3}{2\lambda} W_{-1} \left(-\frac{2\lambda}{3} \left(\frac{C}{y} \right)^{2/3} \right). \quad (9.7)$$

The Lambert representation uses the standard real branches described in [3]. The branch W_{-1} is mandatory: its value tends to $-\infty$ as the argument approaches 0 from below, and therefore $x_0(y) \rightarrow +\infty$. The principal branch W_0 would give the irrelevant small solution.

9.3 All algebraic corrections around the carrier

Write

$$\nu(y) = x_0(y) + D(x_0(y)^{-1}), \quad D(t) = \sum_{m \geq 1} d_m t^m. \quad (9.8)$$

Subtracting the carrier equation from (9.4) gives an exact formal correction equation.

Theorem 9.1 (Lambert-carrier correction equation). *The series $D(t) \in t\mathbb{R}[[t]]$ is the unique formal solution of*

$$\lambda D(t) - p \log(1 + tD(t)) + H\left(\frac{t}{1 + tD(t)}\right) = 0. \quad (9.9)$$

Its coefficients are triangularly determined by

$$d_m = [t^m] \left\{ \frac{p}{\lambda} \log(1 + tD_{<m}(t)) - \frac{1}{\lambda} H\left(\frac{t}{1 + tD_{<m}(t)}\right) \right\}, \quad (9.10)$$

where $D_{<m}(t) = \sum_{j=1}^{m-1} d_j t^j$.

Proof. Put $x = x_0 + D$ in (9.4) and subtract $\lambda x_0 - p \log x_0 = \log(y/C)$. Since $t = x_0^{-1}$,

$$0 = \lambda D - p \log \left(1 + \frac{D}{x_0} \right) + H \left(\frac{1}{x_0 + D} \right),$$

which is (9.9). Its linear coefficient in D at $t = 0$ is $\lambda \neq 0$, so the formal implicit-function theorem gives a unique solution in $t\mathbb{R}[[t]]$. Moreover, the coefficient of t^m on the right side of the rearranged equation

$$D = \frac{p}{\lambda} \log(1 + tD) - \frac{1}{\lambda} H \left(\frac{t}{1 + tD} \right)$$

depends only on $d_1, \dots, d_{m-1}$, proving the recursion. $\square$

9.4 Bell-polynomial form of the inverse coefficients

Expanding the logarithm and the negative powers in (9.10) yields a fully finite formula. With the convention $\widehat{B}_{0,0} = 1$ and impossible indices equal to zero,

$$d_m = \frac{p}{\lambda} \sum_{r=1}^m \frac{(-1)^{r-1}}{r} \widehat{B}_{m-r,r}(d_1, d_2, \dots) \quad (9.11)$$

$$- \frac{1}{\lambda} \sum_{k=1}^m h_k \sum_{r=0}^{m-k} (-1)^r \binom{k+r-1}{r} \widehat{B}_{m-k-r,r}(d_1, d_2, \dots). \quad (9.12)$$

Although d_m appears in the notation on the right, the index constraints ensure that only $d_1, \dots, d_{m-1}$ occur.

The first coefficients are

$$d_1 = -\frac{h_1}{\lambda}, \quad (9.13)$$

$$d_2 = -\frac{ph_1 + \lambda h_2}{\lambda^2}, \quad (9.14)$$

$$d_3 = -\frac{\lambda h_1^2 + p^2 h_1 + p\lambda h_2 + \lambda^2 h_3}{\lambda^3}, \quad (9.15)$$

$$d_4 = -\frac{1}{2\lambda^4} (5p\lambda h_1^2 + 6\lambda^2 h_1 h_2 + 2p^3 h_1 + 2p^2 \lambda h_2 + 2p\lambda^2 h_3 + 2\lambda^3 h_4). \quad (9.16)$$

Numerically,

$$\begin{aligned} d_1 &= -0.09264385352561312984200, \\ d_2 &= -0.5885630399313472628118, \\ d_3 &= -2.596680167407531021207, \\ d_4 &= -13.14905339996706568529, \\ d_5 &= -85.49870151462556652201, \\ d_6 &= -711.262914832946 \dots, \\ d_7 &= -7306.78827028497 \dots, \\ d_8 &= -89561.7608826372 \dots \end{aligned}$$

Therefore the Lambert-normalized inverse is

$$\boxed{\nu(y) \sim x_0(y) + \frac{d_1}{x_0(y)} + \frac{d_2}{x_0(y)^2} + \frac{d_3}{x_0(y)^3} + \dots} \quad (9.17)$$

Theorem 9.2 (Asymptotic validity of the correction series). *For every fixed $M \geq 1$, let*

$$\nu_M(y) := x_0(y) + \sum_{m=1}^{M-1} d_m x_0(y)^{-m}.$$

Then substitution into the logarithmic forward model gives

$$\lambda \nu_M - p \log \nu_M + H(\nu_M^{-1}) - \log(y/C) = \mathcal{O}(x_0^{-M}). \quad (9.18)$$

Since the derivative of the left side with respect to x is $\lambda + \mathcal{O}(x^{-1})$, the corresponding inverse error is

$$\nu(y) - \nu_M(y) = \mathcal{O}(x_0(y)^{-M}). \quad (9.19)$$

Proof. Equation (9.9) determines the coefficient of each power of t uniquely. Truncation after t^{M-1} leaves a residual $\mathcal{O}(t^M)$. The derivative of the logarithmic forward map is bounded away from zero for large x , so the mean-value theorem converts the residual into an inverse error of the same order. $\square$

10 The inverse in polynomial–logarithmic form

The Lambert form is compact and numerically stable. For algebraic manipulation it is useful to eliminate W_{-1} and expand directly in the canonical polynomial–logarithmic scale.

10.1 Normalization

Define

$$X := \frac{1}{\lambda} \log \frac{y}{C}, \quad \ell := \log X, \quad q := \frac{p}{\lambda} = \frac{3}{2\lambda}, \quad \eta_k := \frac{h_k}{\lambda}. \quad (10.1)$$

Then (9.4) becomes

$$x - q \log x + G(x^{-1}) = X, \quad G(u) := \sum_{k \geq 1} \eta_k u^k. \quad (10.2)$$

Seek

$$\boxed{\nu(y) \sim X + q\ell + \sum_{m \geq 1} \frac{P_m(\ell)}{X^m}.} \quad (10.3)$$

Theorem 10.1 (Polynomial–log inverse recursion). *Put $t = X^{-1}$ and $R(t, \ell) = \sum_{m \geq 1} P_m(\ell)t^m$. Then R is the unique formal solution of*

$$\boxed{R - q \log(1 + t(q\ell + R)) + G\left(\frac{t}{1 + t(q\ell + R)}\right) = 0.} \quad (10.4)$$

Consequently

$$\boxed{P_m(\ell) = [t^m] \left\{ q \log(1 + t(q\ell + R_{<m})) - G\left(\frac{t}{1 + t(q\ell + R_{<m})}\right) \right\},} \quad (10.5)$$

where $R_{<m} = \sum_{j=1}^{m-1} P_j(\ell)t^j$. Each P_m is a polynomial of degree m , with leading term

$$P_m(\ell) = \frac{(-1)^{m+1}}{m} q^{m+1} \ell^m + \text{a polynomial of degree at most } m-1. \quad (10.6)$$

Proof. Insert $x = X + q\ell + R = X(1 + t(q\ell + R))$ into (10.2). Since $\log x = \ell + \log(1 + t(q\ell + R))$, the terms $q\ell$ cancel and one obtains (10.4). The same t -adic triangularity used in Theorem 9.1 proves existence, uniqueness, and (10.5). The highest power of ℓ at order t^m comes from the m th term in the logarithm of $1 + tq\ell$ and equals $(-1)^{m+1} q^{m+1} \ell^m / m$. $\square$

10.2 A completely explicit Bell recursion

For $N, r \geq 0$ define

$$\mathcal{P}_{N,r}(\ell) := [t^N] \left(q\ell + \sum_{j \geq 1} P_j(\ell)t^j \right)^r = \sum_{j=0}^{\min(r,N)} \binom{r}{j} (q\ell)^{r-j} \widehat{B}_{N,j}(P_1, P_2, \dots). \quad (10.7)$$

Then (10.5) is equivalent to

$$P_m(\ell) = q \sum_{r=1}^m \frac{(-1)^{r-1}}{r} \mathcal{P}_{m-r,r}(\ell) \quad (10.8)$$

$$- \sum_{k=1}^m \eta_k \sum_{r=0}^{m-k} (-1)^r \binom{k+r-1}{r} \mathcal{P}_{m-k-r,r}(\ell). \quad (10.9)$$

The index bounds again ensure that only $P_1, \dots, P_{m-1}$ appear on the right. Thus (10.8)–(10.9) is a general finite coefficient formula in exactly the ordinary-Bell-polynomial language used for the forward expansion.

10.3 First polynomial coefficients

The first four polynomials are

$$P_1(\ell) = q^2\ell - \eta_1, \quad (10.10)$$

$$P_2(\ell) = -\frac{1}{2}q^3\ell^2 + (\eta_1q + q^3)\ell - \eta_1q - \eta_2, \quad (10.11)$$

$$P_3(\ell) = \frac{1}{3}q^4\ell^3 - \left(\eta_1q^2 + \frac{3}{2}q^4\right)\ell^2 + (3\eta_1q^2 + 2\eta_2q + q^4)\ell - \eta_1^2 - \eta_1q^2 - \eta_2q - \eta_3, \quad (10.12)$$

$$P_4(\ell) = -\frac{1}{4}q^5\ell^4 + \left(\eta_1q^3 + \frac{11}{6}q^5\right)\ell^3 + \left(-\frac{11}{2}\eta_1q^3 - 3\eta_2q^2 - 3q^5\right)\ell^2 + (3\eta_1^2q + 6\eta_1q^3 + 5\eta_2q^2 + 3\eta_3q + q^5)\ell - \frac{5}{2}\eta_1^2q - 3\eta_1\eta_2 - \eta_1q^3 - \eta_2q^2 - \eta_3q - \eta_4. \quad (10.13)$$

For the rooted-tree constants,

$$\begin{aligned} q &= 1.38407334242790228719882463131\dots, \\ \eta_1 &= 0.09264385352561312984200\dots, \\ \eta_2 &= 0.46033715192675089877\dots, \\ \eta_3 &= 1.77348286990414903\dots, \\ \eta_4 &= 9.4092963678408456\dots \end{aligned}$$

In particular, the first two scales are

$$\nu(y) = \frac{1}{\lambda} \log \frac{y}{C} + \frac{3}{2\lambda} \log \left(\frac{1}{\lambda} \log \frac{y}{C} \right) + \mathcal{O} \left(\frac{\log \log y}{\log y} \right). \quad (10.14)$$

The complete correction at every order is supplied by [Theorem 10.1](#).

Structural checkpoint

The two inverse forms are equivalent but serve different purposes. Equation (9.17) keeps the entire logarithmic carrier inside W_{-1} and is numerically superior. Equation (10.3) expands the same answer in the canonical field $\mathbb{R}[\log X][[X^{-1}]]$ and exposes every coefficient as a polynomial.

11 The discrete generalized inverse and lattice effects

For large n , a_n is strictly increasing. Define

$$N(y) := \min\{n \geq 1 : a_n \geq y\}. \quad (11.1)$$

The step width on the logarithmic y -axis tends to $\log \alpha$, because $a_{n+1}/a_n \rightarrow \alpha$. Hence no continuous asymptotic series can determine the position inside each unit step.

Theorem 11.1 (Smooth versus discrete inverse). *Let ν be any smooth asymptotic inverse whose forward interpolation matches (6.16) to all algebraic orders. Then*

$$N(y) = \nu(y) + \mathcal{O}(1), \quad y \rightarrow \infty. \quad (11.2)$$

Moreover, at sequence values,

$$\nu(a_n) = n + \mathcal{O}(n^{-M}) \quad (11.3)$$

for every fixed M , provided the forward interpolation is retained to the corresponding order.

Proof. The logarithmic derivative of the dominant interpolation tends to $\lambda > 0$. Multiplying y by a bounded factor therefore shifts its smooth inverse by a bounded amount, while consecutive exact values differ by the asymptotically fixed factor α . This proves (11.2). At $y = a_n$, an M -order relative forward error is $\mathcal{O}(n^{-M})$; division by the nonzero limiting logarithmic derivative preserves that order in the inverse variable. $\square$

A lattice-completed approximation may be written formally as

$$N_{\text{lat}}(y) := \lceil \nu(y) \rceil. \quad (11.4)$$

It is correct away from vanishing neighborhoods of the exact jump values, but no asymptotic formula can choose the value exactly at a discontinuity without specifying the convention in (11.1). The bounded sawtooth $\lceil x \rceil - x$ is therefore a genuine extra sector, not a term of the descending power-logarithmic series.

A point requiring care

Exponentially small changes in a smooth interpolation can move a value across an integer boundary. Thus the algebraic inverse transseries is canonical, but an exact below-unit description of the staircase requires both a chosen interpolation and an explicit rounding convention.

12 Beyond-all-orders coefficient transseries

12.1 Why one dominant power series is not the whole story

The expansion in Theorem 6.1 is complete in the Poincaré sense at the dominant exponential scale ρ^{-n} . It cannot detect a term such as σ^{-n} with $|\sigma| > \rho$, because

$$\frac{|\sigma|^{-n}}{\rho^{-n}} = \left(\frac{\rho}{|\sigma|} \right)^n$$

decays faster than every power of n^{-1} . Such terms are precisely the next transseries sectors.

The recursive equation itself predicts inherited singularities. If T has a singularity at τ , then the summand $T(z^k)/k$ in $R(z)$ has singular germs at every k th root of τ . In addition, even when R is regular, the outer Cayley function becomes critical whenever

$$\zeta(z) = e^{-1} \quad (12.1)$$

on the chosen continued sheet. Thus there are two mechanisms:

1. *inherited preimages*: z^k reaches an earlier singularity;
2. *new Cayley critical points*: the continued analytic input reaches the branch value e^{-1} .

The visible inherited skeleton generated by the principal singularity is

$$\mathcal{R} := \left\{ \rho^{1/m} e^{2\pi i j/m} : m \geq 1, 0 \leq j < m \right\}. \quad (12.2)$$

Its level- m coefficient weights are

$$\sigma_{m,j}^{-n} = \rho^{-n/m} e^{-2\pi i j n/m}. \quad (12.3)$$

For integer n , the phase is periodic of period dividing m . Conjugate singularities combine into real oscillatory sectors.

A point requiring care

Equation (12.2) is a guaranteed recursive skeleton of candidate inherited germs, not by itself an exhaustive global singularity theorem. Analytic continuation may encounter additional critical solutions of (12.1), and a critical point may coalesce with an inherited germ. The local calculus below handles all of these cases once the relevant continued germs have been identified.

12.2 General transfer from a singular germ

Let $\sigma \neq 0$ be an accessible singularity on a specified sheet, and put $u = 1 - z/\sigma$. Suppose its nonanalytic local part has a generalized Puiseux expansion

$$T(z) \sim A_\sigma(u) + \sum_{\beta \in E_\sigma} c_{\sigma,\beta} u^\beta, \quad (12.4)$$

where A_σ is analytic, E_σ is locally finite and well ordered by real part, and no displayed β is a nonnegative integer.

Theorem 12.1 (Universal singular-sector coefficient formula). *Under the usual dented-domain condition at σ , the contribution of (12.4) to the coefficient transseries is*

$$\mathfrak{S}_\sigma(n) \sim \sigma^{-n} \sum_{\beta \in E_\sigma} \frac{c_{\sigma,\beta}}{\Gamma(-\beta)} n^{-\beta-1} \sum_{r \geq 0} \frac{g_r(\beta)}{n^r}. \quad (12.5)$$

Thus the coefficient multiplying $\sigma^{-n} n^{-\beta-1-r}$ is exactly

$$A_{\sigma,\beta,r} = \frac{c_{\sigma,\beta}}{\Gamma(-\beta)} g_r(\beta). \quad (12.6)$$

Proof. Apply the exact transfer identity (6.1), with ρ replaced by σ , to every noninteger monomial in the local germ, and then use the gamma-ratio expansion (2.9). Analytic powers belong to the regular local part and do not contribute an infinite large- n tail. Hankel-contour deformation justifies summing any prescribed finite truncation under the dented-domain hypothesis. $\square$

This theorem is the full analogue of (6.5); it allows arbitrary ramification exponents, not only half-integers.

12.3 A finite-radius transseries theorem

Fix R with $\rho < R < 1$. The following hypothesis is explicit enough for a rigorous finite-level statement.

Definition 12.2 (Continuation hypothesis $H(R)$). We say that $H(R)$ holds if a chosen continuation of T is analytic in $|z| < R$ after removing finitely many nonintersecting cuts beginning at a finite set Σ_R , every $\sigma \in \Sigma_R$ has a generalized Puiseux germ of the form (12.4), the germs are Delta-regular, and the remainder of the deformed coefficient contour lies on a circle of radius strictly larger than every retained singularity and at least R up to harmless indentations.

Theorem 12.3 (Finite-level coefficient transseries). Assume $H(R)$. Order Σ_R by increasing modulus. For any finite truncation of the exponent and inverse-power sums,

$$a_n \sim \sum_{\sigma \in \Sigma_R} \mathfrak{S}_\sigma(n) + \mathcal{O}(R^{-n} n^K) \quad (12.7)$$

for a finite exponent K determined by the local remainder bounds. Every coefficient in every retained sector is given by (12.6).

Proof. Deform the Cauchy coefficient contour into the sum of small Hankel contours around the cuts issuing from points in Σ_R plus an outer remainder contour. Apply Theorem 12.1 on each Hankel contour. The outer contour is exponentially bounded by R^{-n} times a polynomial factor coming from the allowed boundary growth. Since only finitely many singularities and local terms are retained, the deformation and summation are legitimate term by term. $\square$

When the level- m inherited points are simple square-root germs, their combined contribution has the particularly transparent form

$$\rho^{-n/m} n^{-3/2} \sum_{j=0}^{m-1} e^{-2\pi i j n/m} \sum_{r \geq 0} \frac{A_{m,j,r}}{n^r}. \quad (12.8)$$

Relative to the dominant sector, its exponential smallness is

$$\exp \left[-\lambda \left(1 - \frac{1}{m} \right) n \right]. \quad (12.9)$$

The distinction between the Poincaré sector and exponentially small sectors follows the transseries conventions emphasized in [13]. The first inherited level, $m = 2$, is therefore of relative order $e^{-\lambda n/2}$, beyond every inverse power of n .

12.4 Recursive construction of every local germ

The singularity recursion can be implemented entirely with the same Bell polynomials used earlier. Let $z = \sigma(1 - u)$. If $\tau = \sigma^k$ is a parent singularity, then

$$1 - \frac{z^k}{\tau} = 1 - (1 - u)^k = ku \left(1 + \sum_{r \geq 1} p_{k,r} u^r \right), \quad p_{k,r} = \frac{(-1)^r}{k} \binom{k}{r+1}. \quad (12.10)$$

For a parent monomial $(1 - z^k/\tau)^\beta$,

$$\left(1 - \frac{z^k}{\tau} \right)^\beta = (ku)^\beta \sum_{N \geq 0} u^N \sum_{j=0}^N \binom{\beta}{j} \widehat{B}_{N,j}(p_{k,1}, p_{k,2}, \dots). \quad (12.11)$$

Thus every inherited local coefficient is a finite ordinary-Bell sum of parent coefficients.

After all singular summands $T(z^k)/k$ have been propagated, add the regular Taylor parts to obtain the local generalized series for $R(z)$. Exponentiate it to form $\zeta(z) = ze^{R(z)}$; after replacing u by a common ramified coordinate $w = u^{1/L}$, this exponentiation is exactly (2.3).

There are then two outer-composition cases.

Noncritical inherited point. If $\zeta(\sigma) \neq e^{-1}$ and the selected Cayley branch is regular there, write $\Delta(u) = \zeta(z) - \zeta(\sigma)$. Then

$$T(z) = \sum_{r \geq 0} \frac{\mathcal{C}^{(r)}(\zeta(\sigma))}{r!} \Delta(u)^r, \quad (12.12)$$

and powers of Δ are computed by ordinary Bell polynomials. The Cayley derivatives themselves may be generated from

$$\mathcal{C}'(w) = \frac{\mathcal{C}(w)}{w(1 - \mathcal{C}(w))} \quad (12.13)$$

by differentiation and Faà di Bruno calculus.

Critical point. If $\zeta(\sigma) = e^{-1}$, define $q_\sigma(u) = 1 - e\zeta(z)$. Compose its leading ramified series with the universal Cayley germ

$$T(z) = 1 + \sum_{r \geq 1} c_r q_\sigma(u)^{r/2}. \quad (12.14)$$

The coefficient extraction is the ramified version of (5.7). If the leading exponent of q_σ is δ , then a generic critical composition begins with exponent $\delta/2$. Hence a coincidence between an inherited square root and a Cayley critical point can create quarter powers; [Theorem 12.1](#) already allows this.

Algorithm 12.4 (Local-to-global transseries engine). To compute every sector inside a prescribed radius $R < 1$ and to a prescribed power order:

1. Enumerate accessible inherited preimages and continued critical solutions in $|z| < R$, recording the parent relation and branch.
2. At each point σ , choose a common ramification index L and propagate all parent germs by (12.11).
3. Sum the propagated germs in $R(z)$ and exponentiate with complete Bell polynomials to obtain the local germ of ζ .
4. Use (12.12) or (12.14) to obtain the local germ of T .
5. Transfer every noninteger monomial with (12.6); pair conjugate sectors to obtain a manifestly real expression on integer n .

Every operation at a fixed truncation order is finite and triangular.

Beyond-all-orders completion

The all-orders dominant sector is completely explicit and unconditional. The recursive engine in [Theorem 12.4](#) is a constructive beyond-all-orders completion: once a finite set of continued singular germs has been certified, every coefficient in their exponential sectors follows from finite Bell-polynomial operations. A natural next theorem would prove $H(R)$ uniformly as $R \uparrow 1$ and classify all critical coincidences on the resulting Riemann surface.

13 Beyond-all-orders inversion

13.1 A chosen transseries interpolation

For noninteger x , an oscillatory factor σ^{-x} requires a branch of $\text{Log } \sigma$. Different branches agree at integer $x = n$ but generally differ between integers. Therefore the exponentially small part of a *functional* inverse is not canonical until a continuation convention has been chosen.

Fix such a convention and pair conjugate branches so that the interpolation is real on the positive real axis. Write the completed forward transseries as

$$\mathfrak{a}(x) = \mathcal{A}_0(x)(1 + \mathcal{E}(x)), \quad \mathcal{A}_0(x) = C e^{\lambda x} x^{-p} S(x^{-1}), \quad (13.1)$$

where $\mathcal{E}$ is a sum of exponentially small sectors. A term from a singularity $\sigma = |\sigma|e^{i\theta}$ and local exponent β has, relative to $\mathcal{A}_0$, the scale

$$Q_{\sigma,\beta}(x) := \exp[-\mu_\sigma x - i\theta x] x^{p-\beta-1}, \quad \mu_\sigma := \log \frac{|\sigma|}{\rho} > 0, \quad (13.2)$$

multiplied by a descending power series. For $\sigma = \rho^{1/m} e^{2\pi i j/m}$ and a square-root exponent $\beta = 1/2$, this reduces to

$$Q_{m,j}(x) = e^{-\lambda(1-1/m)x} e^{-2\pi i j x/m}, \quad (13.3)$$

with no extra leading power of x .

13.2 Carrier-first implicit equation

Let $\nu_0(y)$ be the complete algebraic inverse from [Section 9](#); thus $\mathcal{A}_0(\nu_0) = y$. Put

$$\nu_{\text{full}}(y) = \nu_0(y) + \delta(y). \quad (13.4)$$

With

$$F_0(x) := \log \mathcal{A}_0(x) = \log C + \lambda x - p \log x + H(x^{-1}), \quad (13.5)$$

the exact transseries correction equation is

$$\boxed{F_0(\nu_0 + \delta) - F_0(\nu_0) + \log(1 + \mathcal{E}(\nu_0 + \delta)) = 0.} \quad (13.6)$$

The derivative

$$F'_0(x) = \lambda - \frac{p}{x} - \frac{H'(x^{-1})}{x^2} = \lambda + \mathcal{O}(x^{-1}) \quad (13.7)$$

is invertible in the slow coefficient field.

Theorem 13.1 (General coefficient recursion for the inverse transseries). *Choose finitely many independent small transmonomials $\mathbf{Q} = (Q_1, \dots, Q_s)$ and truncate the forward transseries accordingly. There is a unique formal solution*

$$\delta = \sum_{\mathbf{n} \in \mathbb{N}_0^s \setminus \{\mathbf{0}\}} D_{\mathbf{n}}(\nu_0^{-1}, \log \nu_0) \mathbf{Q}^{\mathbf{n}} \quad (13.8)$$

of (13.6), ordered by total transmonomial valuation. Its coefficients obey the triangular recursion

$$\boxed{D_{\mathbf{n}} = -\frac{1}{F'_0(\nu_0)}[\mathbf{Q}^{\mathbf{n}}] \left\{ \sum_{r \geq 2} \frac{F_0^{(r)}(\nu_0)}{r!} \delta^r + \log(1 + \mathcal{E}(\nu_0 + \delta)) \right\}}, \quad (13.9)$$

where the coefficient on the right is evaluated using only terms of δ of strictly smaller transmonomial valuation. Every $D_{\mathbf{n}}$ then has a computable descending power-logarithmic expansion in ν_0 .

Proof. Expand the first term of (13.6) at ν_0 :

$$F_0(\nu_0 + \delta) - F_0(\nu_0) = F'_0(\nu_0)\delta + \sum_{r \geq 2} \frac{F_0^{(r)}(\nu_0)}{r!} \delta^r.$$

Since $F'_0(\nu_0)$ has nonzero leading term λ , it is invertible. At any nonzero multiindex, the coefficient of the linear term is $F'_0(\nu_0)D_n$. Every other occurrence of δ is either nonlinear or multiplied by a positive transmonomial from $\mathcal{E}$, so it contains only lower coefficients in the well-ordered support. Solving the linear equation gives (13.9). The slow-field expansion uses the same Bell and composition recursions as Section 10. $\square$

The first exponential correction is especially simple. If $\mathcal{E}^{[1]}$ denotes the part linear in the chosen small monomials, then

$$\delta^{[1]}(y) = -\frac{\mathcal{E}^{[1]}(\nu_0(y))}{F'_0(\nu_0(y))}. \quad (13.10)$$

Higher sectors include products of forward sectors, derivatives caused by the shift $x \mapsto x + \delta$, and the logarithmic cumulants of $1 + \mathcal{E}$; (13.9) accounts for all of them without case-by-case guessing.

13.3 Canonical and noncanonical information

The inverse hierarchy has three layers.

1. The Lambert carrier and its algebraic corrections are canonical and depend only on the dominant Poincaré sector.
2. Exponentially small smooth corrections are canonical only after the analytic continuation and logarithm branches defining the interpolation have been fixed.
3. The exact discrete generalized inverse additionally requires the lattice rounding described in Section 11.

This hierarchy is not a defect. It records exactly how much information the integer sequence itself determines at each asymptotic scale.

14 Coefficient algorithms and reproducibility

14.1 Exact sequence generation

The following sieve form of (3.5) avoids recomputing divisors.

Listing 1: Exact generation of $a_1, \dots, a_N$ (language-neutral pseudocode).

```

a[0] := 0
b[0..N] := 0
a[1] := 1
for multiple = 1..N:
    b[multiple] += 1           # contribution 1*a[1]

for n = 2..N:
    numerator := sum_{k=1}^{n-1} b[k] * a[n-k]
    assert numerator mod (n-1) = 0
    a[n] := numerator / (n-1)
    for multiple = n, 2n, 3n, ... <= N:
        b[multiple] += n * a[n]
```

The integer divisibility assertion is a useful exact check on every newly computed term.

14.2 Solving for ρ without evaluating T at its boundary

Equation (3.7) uses only $T(r^k)$ with $k \geq 2$. Even when r is extremely close to ρ , these arguments remain uniformly subcritical. A certified procedure is therefore:

1. generate enough exact a_n by Listing 1;
2. evaluate each $T(r^k)$ by a truncated positive series with a tail majorant;
3. use interval Newton or bisection on $f(r) = \log r + 1 + \sum_{k \geq 2} T(r^k)/k$;
4. verify $f'(r) > 0$ on the final interval.

This is much more stable than attempting to estimate the singularity from ratios a_{n+1}/a_n , whose convergence carries inverse-power corrections.

14.3 Power, exponential, and logarithm recurrences

Bell polynomials are conceptually clean, but dense recurrences are faster in software. If

$$F(u) = 1 + \sum_{n \geq 1} f_n u^n, \quad F(u)^\gamma = \sum_{n \geq 0} v_n u^n,$$

then differentiation of $Fv' = \gamma v F'$ gives

$$v_0 = 1, \quad v_n = \frac{1}{n} \sum_{k=1}^n ((\gamma + 1)k - n) f_k v_{n-k}. \quad (14.1)$$

If $A(u) = \sum_{n \geq 0} A_n u^n$ and $E(u) = e^{A(u)} = \sum E_n u^n$, then

$$E_0 = e^{A_0}, \quad E_n = \frac{1}{n} \sum_{k=1}^n k A_k E_{n-k}. \quad (14.2)$$

If $F(0) = 1$ and $H = \log F$, then

$$H_n = f_n - \frac{1}{n} \sum_{k=1}^{n-1} k H_k f_{n-k}. \quad (14.3)$$

These recurrences are exactly equivalent to the Bell-polynomial formulas and are convenient for high-order numerical work.

14.4 End-to-end dominant-sector pipeline

Algorithm 14.1 (All coefficients through order M). Given a target order M :

1. Generate exact tree numbers and solve (3.7) for ρ .
2. Compute the coefficients of $R(\rho(1-s))$ through s^{2M+1} using (5.16); exponentiate by (14.2) to obtain $q_1, \dots, q_{2M+1}$.
3. Compute $c_1, \dots, c_{2M+1}$ from (4.8), using (14.1) for $\Phi^{-r/2}$.
4. Compute $t_1, \dots, t_{2M+1}$ by (5.6).
5. Compute $g_r(j+1/2)$ from (2.8) and form $\tau_0, \dots, \tau_M$ by (6.5).
6. Normalize to $s_m = \tau_m/\tau_0$, compute h_m by (7.3), then compute d_m by (9.10) or P_m by (10.5).

At every stage, the coefficient of order m depends only on data of order at most m ; no nonlinear global solve is required after ρ is known.

14.5 Symbolic and numerical audit checks

A robust implementation should verify all of the following independent identities.

1. The exact recurrence reproduces the known initial values $1, 1, 2, 4, 9, 20, \dots$
2. The critical residuals $T(\rho) - 1$ and $e\zeta(\rho) - 1$ vanish within the certified numerical enclosure.
3. The identity $1 - e\zeta(\rho(1 - s)) = q(s)$ holds coefficientwise to the requested order.
4. Substituting the computed t_n into $T = \zeta e^T$ leaves a Puiseux residual beyond the requested order.
5. The two formulas (6.5) and direct expansion of the exact transfer coefficient (6.1) agree.
6. Substitution of the d_m into (9.9), and of the P_m into (10.4), leaves zero through the requested order.

These checks separate derivation errors from loss of numerical precision.

15 Numerical validation

15.1 Forward expansion

Let

$$A_M(n) := \frac{\rho^{-n}}{\sqrt{\pi} n^{3/2}} \sum_{m=0}^{M-1} \frac{\tau_m}{n^m}. \quad (15.1)$$

Table 3 compares $A_M(n)$ with exact integers generated by (3.5). The entries are relative errors $|A_M(n)/a_n - 1|$.

Table 3: Relative errors of the dominant asymptotic expansion.				
n	$M = 1$	$M = 4$	$M = 8$	$M = 12$
20	6.5807×10^{-3}	9.7082×10^{-5}	6.3781×10^{-6}	1.5969×10^{-5}
50	2.2224×10^{-3}	1.9668×10^{-6}	3.1903×10^{-9}	5.8824×10^{-11}
100	1.0554×10^{-3}	1.1313×10^{-7}	1.0129×10^{-11}	9.7958×10^{-15}
200	5.1460×10^{-4}	6.8101×10^{-9}	3.6381×10^{-14}	2.0834×10^{-18}
500	2.0280×10^{-4}	1.7064×10^{-10}	2.2767×10^{-17}	3.2504×10^{-23}

At $n = 20$, twelve terms already pass the least term and perform worse than eight terms. At larger n , the optimal truncation moves outward exactly as expected for a divergent asymptotic expansion.

15.2 Inverse expansion at exact sequence values

For $y = a_n$, define

$$\nu_K(y) := x_0(y) + \sum_{m=1}^K \frac{d_m}{x_0(y)^m}. \quad (15.2)$$

Table 4 records the absolute index error $|\nu_K(a_n) - n|$. The “carrier” column means $K = 0$.

Table 4: Absolute errors of the Lambert-carrier inverse.

n	carrier	$K = 1$	$K = 4$	$K = 8$	$K = 12$
20	6.5450×10^{-3}	1.9143×10^{-3}	3.7564×10^{-5}	9.3870×10^{-6}	1.7523×10^{-5}
50	2.1114×10^{-3}	2.5861×10^{-4}	3.3169×10^{-7}	9.9830×10^{-10}	2.7265×10^{-11}
100	9.8802×10^{-4}	6.1593×10^{-5}	9.3442×10^{-9}	1.5282×10^{-12}	2.1783×10^{-15}
200	4.7827×10^{-4}	1.5047×10^{-5}	2.7890×10^{-10}	2.7117×10^{-15}	2.2840×10^{-19}
500	1.8766×10^{-4}	2.3752×10^{-6}	2.7824×10^{-12}	6.7465×10^{-19}	1.4159×10^{-24}

The inverse error is approximately the logarithmic forward residual divided by λ , in agreement with (9.19). The same least-term behavior is visible at $n = 20$.

15.3 Independent recovery of the Puiseux coefficients

A direct implementation of (5.16) with exact a_n , followed by (14.2), (4.8), and (5.6), gives

$$t_1 = -1.55949002037464088554220739589 \dots,$$

$$t_9 = -0.1952123835975564636127699 \dots, \quad t_{17} = -0.1445471935709097054769461 \dots$$

Substitution into (6.5) gives, independently,

$$\tau_0 = 0.7797450101873204427711037 \dots, \quad \tau_8 = 67342.06920114653048340563 \dots$$

These values agree with the published coefficient tables [6] to the accuracy justified by their input precision; the displayed values here were recomputed with higher internal precision.

16 Further deductions and research directions

16.1 Large-order behavior of τ_m

The data strongly suggest factorial-over-power growth of the algebraic coefficients. A precise late-term law should be controlled by the nearest singular obstruction of the local $s^{1/2}$ expansion and, after coefficient transfer, by the first adjacent exponential sectors. The singularity engine of Theorem 12.4 gives a concrete route to this law: compute the first continued germs, derive their Stokes constants relative to the dominant germ, and match the large- m growth of τ_m . This would turn the present finite-level transseries into a quantitative resurgence statement.

16.2 A root-sector conjecture

Conjecture 16.1 (Generic inherited half-power skeleton). *On noncritical continued sheets, every point $\rho^{1/m} e^{2\pi i j/m}$ that is reached without an earlier obstruction carries a square-root germ inherited from $z^m = \rho$. Its coefficient sector therefore has the form (12.8). Critical coincidences are isolated and are the only source of finer ramification on the inherited skeleton.*

This conjecture is local and testable: interval analytic continuation along a prescribed path can certify both accessibility and the noncritical condition $\zeta(\sigma) \neq e^{-1}$.

16.3 Stokes data and lateral continuation

The positive singularity ρ lies on the natural direction of coefficient extraction. To expose sectors farther along related cuts one should compare upper and lower lateral continuations. Their difference is a discontinuity transseries whose coefficients can be propagated by the same local Bell calculus. Pairing lateral values is also the natural way to obtain real median interpolations for the inverse problem.

16.4 Formalization targets

The development separates into layers suitable for mechanization.

1. Purely algebraic identities: ordinary and exponential Bell polynomials, the Cayley coefficient formula, gamma-ratio recurrences, and the triangular inverse recursions.
2. Local analytic theorems: implicit square-root inversion, convergence of the subcritical derivative sums, and fixed-order transfer.
3. Global analytic certificates: continuation domains, singularity enumeration, and contour deformation for $H(R)$.

The first layer is especially well suited to exact rational and algebraic computation, while the second can use interval bounds around numerical constants such as ρ .

17 Conclusion

The Butcher–Pólya rooted-tree numbers admit a particularly clean all-orders calculus once the generating function is factored through the Cayley tree function. The complete chain is

$$\text{subcritical values of } T \longrightarrow \zeta^{(m)}(\rho) \longrightarrow q_m \longrightarrow t_m \longrightarrow \tau_m \longrightarrow h_m \longrightarrow d_m \text{ or } P_m.$$

Every arrow is given by a finite Bell-polynomial or Bernoulli-polynomial formula. The dominant result is the compact convolution (6.5); the functional inverse is the exact Lambert carrier (9.7) with the all-orders correction (9.10), or equivalently the polynomial–logarithmic recursion (10.5).

Beyond all powers, the recursive generating equation supplies a singularity grammar. The universal sector formula (12.6) and the local propagation algorithm turn any certified finite continuation into a coefficient transseries, while (13.9) reverses that transseries around the fully normalized carrier. The remaining global frontier is no longer a question of missing coefficient algebra; it is the analytic classification of accessible sheets, critical preimages, and their Stokes connections.

A Analytic details at the dominant singularity

This appendix records the local analytic argument in a form that makes clear which properties are specific to Pólya trees and which are universal.

Proposition A.1 (Analyticity of the perturbation). *If ρ is the radius of convergence of T , then*

$$R(z) = \sum_{k \geq 2} \frac{T(z^k)}{k}$$

is analytic in $|z| < \sqrt{\rho}$. On every compact subdisc, the series and all of its derivatives converge normally.

Proof. Fix $r < \sqrt{\rho}$. Then $r^2 < \rho$. Since $T(w) = w + \mathcal{O}(w^2)$ at the origin, there are constants M and $r_0 < 1$ such that $|T(w)| \leq M|w|$ for $|w| \leq r_0$. Split the sum into finitely many k for which $r^k > r_0$ and the remaining tail. In the tail,

$$\left| \frac{T(z^k)}{k} \right| \leq \frac{Mr^k}{k},$$

which is summable. The same argument after differentiating uses a polynomial factor in k multiplied by r^{k-m} , still summable. Weierstrass normal convergence proves analyticity and termwise differentiation. $\square$

Proposition A.2 (Critical endpoint and transversality). *Along the positive axis, $\zeta(r)$ is strictly increasing. At the first singularity $r = \rho$,*

$$\zeta(\rho) = e^{-1}, \quad T(\rho) = 1, \quad \zeta'(\rho) > 0.$$

Consequently $q_1 = e\rho\zeta'(\rho) > 0$.

Proof. The coefficients of T , hence of R and $\zeta = ze^R$, are nonnegative; the coefficient of z in ζ is positive. Therefore $\zeta'(r) > 0$ for $r > 0$ in its interval of analyticity. While $\zeta(r) < e^{-1}$, the principal Cayley function $\mathcal{C}(\zeta(r))$ is analytic and solves the functional equation. If ζ did not reach e^{-1} at ρ , the composition $T = \mathcal{C} \circ \zeta$ would extend analytically beyond ρ , contradicting the definition of the radius. Thus $\zeta(\rho) = e^{-1}$, and (4.1) gives $T(\rho) = 1$. Strict positivity of ζ' gives the last assertion. $\square$

Proposition A.3 (Uniqueness on the dominant circle). *The positive point $z = \rho$ is the only singularity of the branch defined at zero on $|z| = \rho$.*

Proof. The series $\zeta(z)$ has nonnegative coefficients. Its first relevant terms are

$$\zeta(z) = z + \frac{1}{2}z^3 + \frac{1}{3}z^4 + \dots,$$

because $R(z) = z^2/2 + z^3/3 + \dots$. For $|z| = \rho$, the triangle inequality gives $|\zeta(z)| \leq \zeta(\rho) = e^{-1}$. Equality would require the nonzero terms z, z^3, z^4 to have the same argument. Writing $z = \rho e^{i\theta}$, this implies $e^{2i\theta} = e^{3i\theta} = 1$, hence $e^{i\theta} = 1$ and $z = \rho$. At every other point the inequality is strict, so the principal Cayley function is analytic at $\zeta(z)$. The perturbation ζ is analytic on the larger disc $|z| < \sqrt{\rho}$ by Theorem A.1; therefore $T = \mathcal{C} \circ \zeta$ is analytic at every other point of $|z| = \rho$. $\square$

Theorem A.4 (Local square-root expansion with analytic remainder). *There is a dented neighborhood of ρ in which*

$$T(z) = 1 + \sum_{n=1}^N t_n \left(1 - \frac{z}{\rho}\right)^{n/2} + \mathcal{O}\left(\left|1 - \frac{z}{\rho}\right|^{(N+1)/2}\right)$$

for every $N \geq 1$, with coefficients given by Theorem 5.1.

Proof. By Theorem A.1, $q(s)$ in (5.2) is analytic near $s = 0$; by Theorem A.2, it has a simple zero with positive coefficient q_1 . Therefore $q(s) = q_1 s(1 + U(s))$, where U is analytic and $U(0) = 0$. Choose a slit neighborhood of $s = 0$ and a consistent square root. The convergent Cayley Puiseux series (4.7) may be composed with this analytic map. Ordinary convergence of the composition and Taylor's theorem give the displayed remainder. Coefficient extraction in the convergent composition is exactly (5.7). $\square$

Together, Theorems A.3 and A.4 supply the standard Delta-domain hypotheses used in the proof of Theorem 6.1.

B Coefficient tables for inversion

Table 5: Normalized forward, logarithmic, and Lambert-correction coefficients.

m	s_m	h_m	d_m
1	0.1004034800964014348	0.1004034800964014348	−0.09264385352561312984
2	0.5039343151023035331	0.4988938856945692936	−0.5885630399313472628
3	1.972284908382278469	1.922025533841821822	−2.596680167407531021
4	10.51734090413157399	10.19739642337376887	−13.14905339996706569
5	73.47647498402047106	71.47146706207073604	−85.49870151462556652
6	645.1397552456612928	630.8599376073372479	−711.2629148329465609
7	6873.530274463408620	6753.629059387619229	−7306.788270284967257
8	86364.21948371140798	85171.04383262353790	−89561.76088263724860
9	$1.250954458605978292 \times 10^6$	$1.236987135642947527 \times 10^6$	$-1.274829503083708903 \times 10^6$
10	$2.051559456480361859 \times 10^7$	$2.032577710708850382 \times 10^7$	$-2.064109900032138585 \times 10^7$

The rapid growth does not contradict asymptotic validity. The series are to be truncated according to the large parameter, and the usable order grows with n or x_0 .

C Derivation of selected inverse coefficients

For illustration, expand (9.9) through t^4 . Since $D = d_1t + d_2t^2 + d_3t^3 + d_4t^4 + \dots$, one has

$$\begin{aligned}
 \log(1 + tD) &= d_1t^2 + d_2t^3 + \left(d_3 - \frac{1}{2}d_1^2\right)t^4 + \mathcal{O}(t^5), \\
 \frac{t}{1 + tD} &= t - d_1t^3 - d_2t^4 + \mathcal{O}(t^5), \\
 H\left(\frac{t}{1 + tD}\right) &= h_1t + h_2t^2 + (h_3 - h_1d_1)t^3 \\
 &\quad + (h_4 - 2h_2d_1 - h_1d_2)t^4 + \mathcal{O}(t^5).
 \end{aligned}$$

Equating powers of t gives

$$\begin{aligned}
 \lambda d_1 + h_1 &= 0, \\
 \lambda d_2 - p d_1 + h_2 &= 0, \\
 \lambda d_3 - p d_2 + h_3 - h_1 d_1 &= 0, \\
 \lambda d_4 - p \left(d_3 - \frac{1}{2}d_1^2\right) + h_4 - 2h_2d_1 - h_1d_2 &= 0.
 \end{aligned}$$

Successive substitution yields (9.13)–(9.16). This small calculation displays the triangular mechanism that the general Bell formula (9.11)–(9.12) packages at arbitrary order.

For completeness, the next coefficient is

$$\begin{aligned}
 d_5 = -\frac{1}{2\lambda^5} &\left(4\lambda^2h_1^3 + 9\lambda p^2h_1^2 + 14\lambda^2ph_1h_2 + 8\lambda^3h_1h_3 + 2p^4h_1 \right. \\
 &\quad \left. + 4\lambda^3h_2^2 + 2\lambda p^3h_2 + 2\lambda^2p^2h_3 + 2\lambda^3ph_4 + 2\lambda^4h_5\right). \tag{C.1}
 \end{aligned}$$

D A compact formula sheet

For reference, the complete dominant and inverse pipeline is collected here.

$$\begin{aligned}
T(z) &= z \exp \left(\sum_{k \geq 1} \frac{T(z^k)}{k} \right), \\
\log \rho + 1 + \sum_{k \geq 2} \frac{T(\rho^k)}{k} &= 0, \\
q_m &= \frac{(-1)^{m+1} e \rho^m}{m!} \zeta^{(m)}(\rho), \\
c_r &= -\frac{2^{r/2}}{r} [v^{r-1}] \left(\frac{2(1 - (1-v)e^v)}{v^2} \right)^{-r/2}, \\
t_n &= \sum_{\substack{1 \leq r \leq n \\ r \equiv n \pmod{2}}} c_r q_1^{r/2} [s^{(n-r)/2}] (1 + U(s))^{r/2}, \\
\kappa_m(\beta) &= \frac{(-1)^{m+1}}{m(m+1)} (B_{m+1}(-\beta) - B_{m+1}(1)), \\
g_r(\beta) &= \frac{1}{r!} Y_r(1! \kappa_1, \dots, r! \kappa_r), \\
\tau_m &= \sum_{j=0}^m \frac{(-1)^{j+1} (2j+1)!!}{2^{j+1}} t_{2j+1} g_{m-j}(j + \tfrac{1}{2}), \\
h_m &= \sum_{k=1}^m \frac{(-1)^{k-1}}{k} \widehat{B}_{m,k}(s_1, s_2, \dots), \\
x_0(y) &= -\frac{3}{2\lambda} W_{-1} \left(-\frac{2\lambda}{3} (C/y)^{2/3} \right), \\
d_m &= [t^m] \left\{ \frac{p}{\lambda} \log(1 + tD_{<m}) - \frac{1}{\lambda} H \left(\frac{t}{1 + tD_{<m}} \right) \right\}, \\
\nu(y) &\sim x_0(y) + \sum_{m \geq 1} d_m x_0(y)^{-m}.
\end{aligned}$$

For a general singular germ $c_{\sigma,\beta}(1 - z/\sigma)^\beta$, replace the dominant formula by the universal sector coefficient

$$A_{\sigma,\beta,r} = \frac{c_{\sigma,\beta}}{\Gamma(-\beta)} g_r(\beta).$$

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